Radio Presets — Recommended Settings for 30 Radios

Recommended **HBC plugin settings** and **radio-side setup** for 20 popular VHF/UHF radios (16 handhelds, 4 mobiles) and 10 HF radios.

How to use this document:

1. Find your radio below (summary table first, details after).
2. Apply its **profile** (A/B/C baseline settings, defined next) in the plugin's Audio Setup tab.
3. Apply the per-radio **overrides and radio-side settings** from its entry.

> These are field-tested starting points, not laws. Two rules are absolute: **every station on a channel must run the same modem and the same channel-access mode/timing**, and on VOX radios **Guard must exceed the slowest radio's VOX hang** (calibration procedure in the ICD §3.6). Menu numbers can shift between firmware revisions — the setting names are what matter.

The three baseline profiles

Profile A — VOX handheld (acoustic coupling or speaker-mic cable)

The zero-hardware path: phone speaker → radio mic, radio keys itself by VOX.

Plugin setting (Audio Setup)	Value
Modem	**OFDM (Rattlegram)** — best survivor of phone/radio audio paths
TX output / PTT	**Speaker — VOX / acoustic**
TX audio stream	**Alarm** (bypasses phone media DSP; set Alarm volume to max)
VOX Lead ms	**250** (covers VOX attack clipping the first tones)
Channel access	Ring for 3+ stations, CSMA for 2
Guard ms	**1500** (covers a fixed ~1–1.5 s VOX hang)
Skip ms	**1200**
Max turn ms	**0** (automatic per modem)

Radio side: VOX ON at medium sensitivity (raise it until data audio keys reliably, no higher), squelch just above the noise, TX time-out (TOT) ≥ 60 s, phone speaker 2–5 cm from the radio mic.

Profile B — Digirig-keyed radio (hard PTT, no VOX)

A Digirig Mobile (CP210x RTS) or Digirig Lite (CM108 GPIO) between phone and radio. The plugin keys PTT electrically — no VOX attack or hang, so turns get much shorter.

Plugin setting (Audio Setup)	Value
Modem	**OFDM** (VHF/UHF)
TX output / PTT	**USB + RTS PTT (Digirig)** — approve the Android USB prompt
TX audio stream	(TX rides the **Media** stream on USB — set Media volume to max)
VOX Lead ms	**0**
Guard ms	**400**
Skip ms	**1200**
Max turn ms	**0** (automatic)

Radio side: VOX OFF, squelch just above noise, TOT ≥ 60 s, mic gain default; start the Digirig attenuation/levels low and raise until decodes are solid (RX audio too hot is the #1 failure).

Profile C — HF SSB (Mercury or PSK31)

Plugin setting (Audio Setup)	Value
Modem	**Mercury HF** (DATAC4, ~87 bps) — or **PSK31** for extreme weak-signal work (very slow: PLI ≈ 16 s of air)
TX output / PTT	**USB + RTS PTT** when the radio/interface gives the plugin a CP210x serial (see entries); otherwise the radio's data-VOX
VOX Lead ms	**0** (hard PTT)
Guard ms	**500**
Skip ms	**1200**
Max turn ms	**0** (automatic; PSK31 sizes its own ~94 s turns)
PLI min s	**120** on HF nets (60 default is busy for DATAC4), with the Ring floor handling the rest

Radio side, always: **USB/upper-sideband data mode** (USB-D / DATA-U / PKT), **speech processor/compressor OFF**, mic EQ off for the data input, power ~25–50 % with **ALC barely moving**, RF gain normal, AGC fast/off per taste, filter wide (≥ 2.4 kHz; the Mercury/OFDM carriers sit near 1500 Hz). **On PSK31, disable or maximize the radio's TX time-out timer — bursts can exceed 60 s.**

> Mercury note: routed Direct messages carry ~29 B of HBC payload on DATAC4 (broadcasts ~39 B) — every standard message type fits.

Summary table

#	Radio	Band	Profile	PTT path
1	Baofeng UV-5R	VHF/UHF HT	A (or B)	VOX / Digirig K1
2	Baofeng BF-F8HP	VHF/UHF HT	A (or B)	VOX / Digirig K1
3	Baofeng UV-82HP	VHF/UHF HT	A (or B)	VOX / Digirig K1
4	Baofeng AR-5RM	VHF/UHF HT	A (or B)	VOX / Digirig K1
5	Quansheng UV-K5(8)	VHF/UHF HT	A (or B)	VOX / Digirig K1
6	TIDRADIO TD-H3	VHF/UHF HT	A (or B)	VOX / Digirig K1
7	Yaesu FT-65R	VHF/UHF HT	A (or B)	VOX / Digirig cable
8	Yaesu FT-60R	VHF/UHF HT	**B**	Digirig (no VOX)
9	Yaesu FT-70DR	VHF/UHF HT	A (or B)	VOX / Digirig cable
10	Yaesu VX-6R	VHF/UHF HT	A	VOX (adapter needed for cable)
11	Icom ID-52A	VHF/UHF HT	**B**	Digirig (treat as no VOX)
12	Icom IC-V86	VHF HT	**B**	Digirig Icom 2-pin (no VOX)
13	Kenwood TH-D75A	VHF/UHF HT	**B**	Digirig K-style 2-pin (no VOX)
14	Anytone AT-D878UVII+	VHF/UHF HT	A (or B)	VOX / Digirig K1
15	Wouxun KG-UV9PX	VHF/UHF HT	A (or B)	VOX / Digirig K1
16	TYT TH-UV88	VHF/UHF HT	A (or B)	VOX / Digirig K1
17	Kenwood TM-V71A	VHF/UHF mobile	**B**	Digirig → 6-pin mini-DIN
18	Icom IC-2730A	VHF/UHF mobile	**B**	Interface → mic jack
19	Yaesu FT-2980R	VHF mobile	**B**	Interface → mic jack
20	Anytone AT-D578UVIII+	VHF/UHF mobile	**B**	Interface → mic jack
21	Icom IC-7300	HF	**C**	Direct USB (codec + CP210x RTS)
22	Icom IC-705	HF–UHF	**C**	Direct USB (codec + CP210x RTS)
23	Icom IC-7100	HF–UHF	**C**	Direct USB (codec + CP210x RTS)
24	Yaesu FT-991A	HF–UHF	**C**	Direct USB (codec + SiLabs RTS)
25	Yaesu FT-891	HF	**C**	Digirig → rear DATA jack
26	Yaesu FT-710	HF	**C**	Direct USB (codec + SiLabs RTS)
27	Yaesu FT-818ND	HF–UHF	**C**	Digirig → 6-pin DATA jack
28	Xiegu G90	HF	**C**	Digirig → 8-pin ACC
29	Xiegu X6100	HF	**C**	USB audio (fw-dependent) or Digirig ACC
30	Kenwood TS-590SG	HF	**C**	Direct USB (codec + SiLabs RTS)

PTT-path legend: "Digirig K1" = Digirig with the Kenwood 2-pin (K1) HT cable. "Direct USB" = phone USB-C OTG straight into the radio's own USB port: the radio's built-in sound card carries audio automatically, and when its internal serial chip is a Silicon Labs CP210x the plugin's **USB + RTS PTT**

keys it directly (set the radio's "USB SEND"/"PTT SELECT"-style menu to **RTS**). If your radio's serial chip is not a CP210x, fall back to the radio's data VOX or a Digirig.

VHF/UHF handhelds

1. Baofeng UV-5R *(reference radio)*

- **Plugin:** Profile A as-is (these defaults were calibrated on this radio). Digirig K1 cable → Profile B.
- **Radio:** `VOX` (menu 4) = 5–7; `SQL` (menu 0) = 3–5; `TOT` (menu 9) ≥ 60 s; `ROGER` OFF; `BCL` OFF; volume ~50–70 % toward the phone mic. VOX hang is fixed ~1–1.5 s — that is what Guard 1500 covers; do not reduce Guard below the measured hang.
- **Notes:** dual-watch (`TDR`) OFF while running data — a reply on the other watch slot garbles RX.

2. Baofeng BF-F8HP

- **Plugin:** Profile A (or B with a K1 Digirig).
- **Radio:** same menu family as the UV-5R: `VOX` menu 4, `SQL` menu 0, `TOT` menu 9 ≥ 60 s, `ROGER`/`BCL` OFF. Use LOW/MID power on the bench — 8 W next to the phone can induce RF into the audio path.
- **Notes:** identical data behavior to the UV-5R; same Guard/Skip.

3. Baofeng UV-82HP

- **Plugin:** Profile A (or B).
- **Radio:** `VOX` menu 4 = 5–7 (VOX keys the *upper* PTT band — make sure the data channel is the A/upper VFO), `SQL` 3–5, `TOT` ≥ 60 s, `ROGER` OFF.
- **Notes:** the dual-PTT button quirk only matters for manual keying.

4. Baofeng AR-5RM

- **Plugin:** Profile A (or B).
- **Radio:** VOX in the menu (level 1–10) = 5; squelch 3–5; TOT ≥ 60 s; roger beep OFF; disable the FM-broadcast radio while running.
- **Notes:** the USB-C port is charge-only — audio still goes through the K1 jack. Spectrum/scan features off during data.

5. Quansheng UV-K5(8)

- **Plugin:** Profile A, but start **VOX Lead 300** — the stock VOX attack clips more than the Baofengs. Digirig K1 → Profile B.

- **Radio:** `VOX` = 5; `SQL` 3–4; `TOT` ≥ 60 s; roger/beep OFF. On modded firmware (IJV/egzumer), keep AGC defaults for data.
- **Notes:** very hot RX audio out — keep volume toward the phone low (30–40 %) to avoid clipping the phone mic.

6. TIDRADIO TD-H3

- **Plugin:** Profile A (or B with a K1 Digirig).
- **Radio:** VOX on (level ~5), squelch 3–5, TOT ≥ 60 s, roger beep and voice prompts OFF.
- **Notes:** ships in "ham/normal/GMRS" modes — any mode works for data; keep the Bluetooth programming link off while operating.

7. Yaesu FT-65R *(reference radio)*

- **Plugin:** Profile A as-is (the second calibration radio). For cabled use: Digirig with the FT-65's 4-conductor 3.5 mm jack cable → Profile B.
- **Radio:** `VOX` (Set Mode item 35) is **ON/OFF only** — no sensitivity adjustment; it keys reliably from speaker audio with the phone close. SQL in Set Mode; `TOT` ≥ 60 s (item "TOT"); beep OFF.
- **Notes:** VOX hang ~1–1.5 s fixed → Guard 1500. The ⓘ dialog in the plugin prints this radio's exact numbers.

8. Yaesu FT-60R

- **Plugin: Profile B** — the FT-60R has **no VOX**; use a Digirig with the 4-conductor 3.5 mm cable (or hold PTT by hand for tests).
- **Radio:** VOX n/a; `TOT` (Set Mode) ≥ 60 s; squelch knob just past noise; beeps OFF.
- **Notes:** a classic, but strictly a cabled/manual-PTT data radio.

9. Yaesu FT-70DR

- **Plugin:** Profile A (VOX exists, adjustable) or B with the 4-conductor cable.
- **Radio:** set VOX in the Set Mode menu (ON, mid sensitivity); force **FM (analog) mode** on the data channel — C4FM/AMS must not engage (AMS OFF); `TOT` ≥ 60 s; beep OFF.
- **Notes:** if AMS flips to digital mid-net, every burst is lost — pin the channel to analog.

10. Yaesu VX-6R

- **Plugin:** Profile A. (Cabled use needs the CT-91/waterproof-jack adapter to split SP/MIC — most owners stay acoustic.)
- **Radio:** `VOX` in the menu (enable, mid gain); `TOT` ≥ 60 s ("TOT" menu); squelch just past noise; beep OFF.
- **Notes:** the screw-down jack is waterproof — hand-tighten after inserting an adapter or the mic line floats.

11. Icom ID-52A

- **Plugin: Profile B** (treat as no usable VOX for data) with an Icom HT cable on the speaker-mic jack; acoustic coupling + manual PTT works for short tests.
- **Radio:** run plain **FM**, not D-STAR (DV OFF); squelch just past noise; `TOT` ≥ 60 s; beeps OFF.
- **Notes:** keep the GPS beacon off — unneeded airtime on the data channel.

12. Icom IC-V86

- **Plugin: Profile B** — no VOX; Digirig with the Icom 2-pin (3.5 mm + 2.5 mm) cable, or acoustic + manual PTT.
- **Radio:** squelch just past noise; `TOT` ≥ 60 s; beep OFF; MID/LOW power on the bench.
- **Notes:** simple, loud 2 m radio — excellent acoustic-RX performer; only TX needs the cable or a thumb.

13. Kenwood TH-D75A

- **Plugin: Profile B** — no VOX; Digirig with the K-style 2-pin cable (the D75's USB-C is programming/TNC, not a sound card for the phone).
- **Radio:** FM mode (not DV); squelch just past noise; TOT ≥ 60 s; turn the **built-in TNC/APRS OFF** so it never keys over HBC bursts.
- **Notes:** speaker output is strong — moderate volume into the Digirig (attenuation on).

14. Anytone AT-D878UVII Plus

- **Plugin:** Profile A (VOX available) or B with a K1 Digirig — prefer B; the DMR-style VOX attack is slow even at max.
- **Radio: analog channel** (not DMR); VOX level 3/3 + VOX delay minimum if going acoustic; squelch 3–5; TOT ≥ 60 s; roger OFF; APRS/GPS beaconing OFF on the data channel.
- **Notes:** if acoustic VOX clips the first symbols, raise plugin VOX Lead to 400 before touching anything else.

15. Wouxun KG-UV9PX

- **Plugin:** Profile A (or B with a K1 Digirig).
- **Radio:** `VOX` menu ON, mid level; squelch 3–5; TOT ≥ 60 s; roger/voice prompts OFF.
- **Notes:** keep the second (dual-watch) band muted or disabled while passing data.

16. TYT TH-UV88

- **Plugin:** Profile A (or B).
- **Radio:** `VOX` menu ON (level ~5), squelch 3–5, TOT ≥ 60 s, beeps OFF.
- **Notes:** behavior tracks the Baofeng family closely; UV-5R values transfer as-is.

VHF/UHF mobiles

17. Kenwood TM-V71A

- **Plugin: Profile B** — Digirig with the TM-V71A 6-pin mini-DIN data cable (PTT is a dedicated pin; RTS keys it cleanly).
- **Radio:** set the data port speed to **1200** (PKT.SPD menu) — that selects the correct (de-emphasized mic-level) audio path for all HBC modems; squelch just past noise; TOT ≥ 60 s (PSK31 excepted — see the Profile C warning).
- **Notes:** audio levels on the 6-pin port are fixed and clean — this is one of the easiest mobiles to interface.

18. Icom IC-2730A

- **Plugin: Profile B** — no data port: use a mic-jack (RJ45) interface cable to the Digirig, with RX audio from the 3.5 mm external-speaker jack.
- **Radio:** squelch just past noise; TOT ≥ 60 s; beeps OFF; LOW/MID power on the bench.
- **Notes:** mic-jack interfacing carries mic gain — keep Digirig TX drive low and bring it up until the far station decodes cleanly with minimal deviation.

19. Yaesu FT-2980R

- **Plugin: Profile B** — mic-jack (RJ45) interface + external speaker jack for RX, keyed by the Digirig.
- **Radio:** squelch just past noise; TOT ≥ 60 s ("TOT" menu); beeps OFF. 80 W is pointless for data across town — run LOW1/LOW2.
- **Notes:** mono-band 2 m; a solid fixed-station data radio once cabled.

20. Anytone AT-D578UVIII Plus

- **Plugin: Profile B** via a mic-jack interface (preferred). Its menu VOX exists but, as on the 878, the attack is slow for data.
- **Radio: analog channel**; squelch 3–5; TOT ≥ 60 s; roger OFF; disable APRS auto-TX on the data channel.
- **Notes:** if you must run VOX, plugin VOX Lead 400 and test carefully for clipped preambles.

HF radios

21. Icom IC-7300

- **Plugin:** Profile C. Phone USB-C (OTG) → radio USB-B directly: the 7300's built-in sound card carries both directions, and its internal **Silicon Labs CP210x** serial lets **USB + RTS PTT** key the radio.

- **Radio:** `SET > Connectors`: **USB SEND = RTS**, DATA MOD = USB, USB AF output ~50 %; mode **USB-D**; speech comp OFF; power so ALC barely moves; TOT long/off for PSK31.
- **Notes:** the single most-popular HF+HBC combination — no Digirig needed at all.

22. Icom IC-705

- **Plugin:** Profile C, direct USB exactly like the 7300 (same codec + CP210x; **USB SEND = RTS**).
- **Radio:** mode USB-D, DATA MOD = USB, comp OFF; its 5–10 W is ample for DATAC4/PSK31. Battery note: USB OTG draws from the phone — carry a powered hub for long field sessions.
- **Notes:** also covers VHF/UHF SSB/FM — the same setup works there.

23. Icom IC-7100

- **Plugin:** Profile C, direct USB (codec + CP210x; set **USB SEND = RTS** in the connectors menu).
- **Radio:** mode USB-D, DATA MOD = USB (ACC/USB setting), comp OFF, ALC barely moving.
- **Notes:** fan runs on TX — normal for long Mercury/PSK31 bursts; keep duty cycle in mind and power at ~30 %.

24. Yaesu FT-991A

- **Plugin:** Profile C, direct USB: built-in codec + Silicon Labs dual COM; set the radio's **PTT SELECT = RTS** (and CAT RTS flow control OFF).
- **Radio:** mode **DATA-USB**; menu `MODE DATA` input = USB; DT GAIN set for barely-moving ALC; speech processor OFF; TOT per Profile C.
- **Notes:** if TX won't key, it is almost always the PTT SELECT menu still on DAKY.

25. Yaesu FT-891

- **Plugin:** Profile C with a **Digirig → rear 6-pin DATA jack** cable (the 891's USB carries CAT only, no audio codec).
- **Radio:** mode **DATA-USB**; menus: DATA PORT SELECT = DATA, DATA PTT SELECT = DAKY (the Digirig keys the DATA jack's PTT pin), DATA OUT LEVEL ~50; comp OFF.
- **Notes:** superb portable match for Mercury; levels via the Digirig pots/Media volume.

26. Yaesu FT-710

- **Plugin:** Profile C, direct USB (built-in codec + SiLabs serial; **PTT SELECT = RTS**).
- **Radio:** mode DATA-U; RPORT/DATA gain for barely-moving ALC; AESS speaker irrelevant for data; comp OFF.
- **Notes:** current-generation sibling of the 7300 workflow.

27. Yaesu FT-818ND

- **Plugin:** Profile C with a **Digirig → 6-pin mini-DIN DATA jack** cable (no USB on the radio).
- **Radio:** mode **DIG** with `DIG MODE = USER-U` (generic USB data); comp n/a; 6 W is fine for DATAC4/PSK31 — run full power with ALC watched.
- **Notes:** the DATA jack expects/returns fixed low-level audio — set phone Media volume max and trim with the Digirig.

28. Xiegu G90

- **Plugin:** Profile C with a **Digirig → 8-pin mini-DIN ACC** cable (no USB audio on the radio).
- **Radio:** mode USB; set input = LINE (ACC); AUX audio levels mid; comp OFF; its 20 W — run 5–10 W for data duty cycle.
- **Notes:** the ACC port carries PTT — the Digirig RTS keys it; no VOX needed.

29. Xiegu X6100

- **Plugin:** Profile C. Recent firmware exposes the radio as a USB sound card — try direct USB first (plugin auto-routes audio; key via the radio's data VOX if its serial is not a CP210x). Otherwise: Digirig on the 3.5 mm ACC jack.
- **Radio:** mode USB; mic/line select = LINE for the ACC path; comp OFF; 5–8 W with ALC barely moving.
- **Notes:** firmware moves fast on this radio — verify audio routing after every update.

30. Kenwood TS-590SG

- **Plugin:** Profile C, direct USB: built-in codec + Silicon Labs virtual COM — enable the radio's **PKT/DATA PTT via RTS** menu so USB + RTS PTT keys it (else use a Digirig on ACC2).
- **Radio:** mode USB + DATA (USB audio input selected in menu); TX EQ/processor OFF for data; power for barely-moving ALC.
- **Notes:** a shack classic; the USB path has been rock-solid for soundcard modes for a decade.

Universal checklist (any radio)

1. Same **modem**, same **channel access**, same **Guard/Skip** on every station.
2. VOX radios: Guard must beat the slowest radio's VOX hang (measure it — ICD §3.6 calibration procedure).
3. Roger beeps, voice prompts, dual-watch, auto-beacons: **OFF** on the data channel.
4. TOT ≥ 60 s for AFSK/OFDM/Mercury; **off or maximum for PSK31**.
5. Acoustic coupling: Alarm volume max, speaker 2–5 cm from mic, quiet room. USB: Media volume max, Digirig levels low-and-raise.
6. Start on LOW power on the bench; RF into the audio path mimics a broken modem.

7. Verify with two stations before the net: a PLI each way, a chat DM each way (watch the delivered/read checkmarks), then a 911 test — that exercises broadcast, direct, ACK and preemption paths.

HBC Radio — KE8TQB — github.com/ke8tqb/HBC-ATAK-Plugin